WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – IV : CHANNELS – SILT THEORIES, WATER LOGGING & CANAL LINING

1. Classification of Canals

> A canal is an artificial channel constructed to carry water from a source (river/ reservoir) to the command area.

❖(a) Based on nature of source

- Permanent (perennial) canals: draw from perennial rivers with head works (weir/ barrage) or reservoirs – water throughout the year (Godavari, Krishna delta canals, Nagarjuna Sagar canals).
- Inundation canals: taken off without head works, flow only when river is in flood (high stage); mostly in Punjab/ Sind historically.

❖(b) Based on function

- Irrigation, carrier, feeder, navigation, power (hydro) and drainage canals.

❖(c) Based on discharge & position in the network

- Main canal (carries entire supply, no direct irrigation – no outlets), branch canals (> 5 cumec), major distributaries (0.25–5 cumec), minor distributaries (< 0.25), water courses (field channels – owned by cultivators).

❖(d) Based on alignment

- Ridge (watershed) canal, contour canal, side-slope canal.

❖(e) Based on lining & construction

- Lined vs unlined; in cutting, in filling, partial cutting & filling.

❖(f) Based on financial output

- Productive (revenue > cost) and protective (famine relief).

2. Canal Alignment

> Alignment = the route/ layout of the canal (plan) selected for economical, efficient command.

❖Alignment types

- Ridge canal (watershed canal): along the ridge line between two drainage basins → commands area on both sides, no cross-drainage works, minimal cutting & filling – best, most economical.
- Contour canal: follows a contour with a small slope, commands area on one side only (down slope); cross-drainage works needed; used in hilly regions.
- Side-slope canal: runs perpendicular to contours (down the slope); no cross-drainage problems, but silting/ scour issues; rare.

❖Requirements for good alignment

- Commanding position (FSL above ground level of the area → gravity flow); shortest length; minimum cross-drainage works; balanced cutting & filling (balancing depth); avoid rocky strata, sandy/ porous soil, villages, valuable land, historic places; minimum curves (radius ≥ 5–6 × bed width); proper drainage & water-logging avoidance; head regulator at suitable place; easy access for maintenance; no unnecessary diversions.

3. Inundation Canals

- No permanent diversion works; water enters when river water level exceeds canal bed; flows during monsoon floods only → cultivation of rabi crops with flood-moisture; silt-laden fertile water deposits.
- Advantages: cheap (no weir), natural silt manuring. Disadvantages: uncertain & seasonal, heavy silting, damage by flood, irregular supply, not for perennial crops.

4. Cross-section of an Irrigation Channel

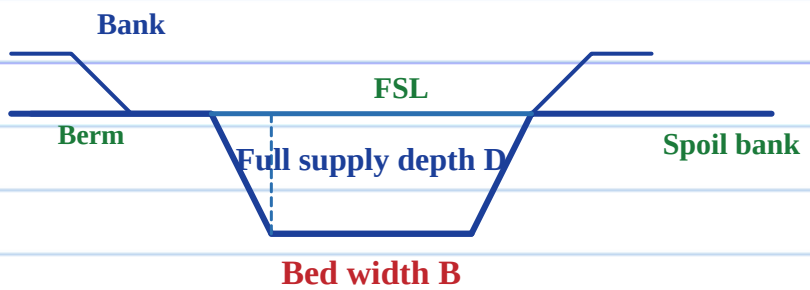


Fig 4.1 Cross-section of an irrigation channel (water level = FSL)

❖Terms

- Full supply level (FSL): water surface in design discharge; full supply depth D; bed width B; bed level; side slopes (cutting 1:1 – 1.5:1 H:V; filling 1.5:1 – 2:1; in sandy 2:1).
- Berm: horizontal strip of land (≥ 1–4 m) left between toe of bank and edge of cut (or borrow pit) for stability, collect scouring, protect bank, space for silt, future widening.
- Freeboard: vertical distance bank top – FSL (0.5 – 1.0 m for small, up to 1.5 m for large canals) – to prevent overtopping.
- Banks: embankments on both sides; top width (service road 5–7.5 m on one side, 3–4 m on the other); inspection road.
- Dowel (Dowla): small bank on road side to protect the canal from rain-cut, 0.3–0.5 m high, 1–2 m wide; backing/ berm.
- Hydraulic gradient line (HGL) & saturation line (the line of seepage through bank; keep it within the bank section at safe slope 4:1 to 6:1).

❖Types of canal section (by relation to ground)

- Fully in cutting (deep, costly; avoids bank), fully in filling (on embankment – seepage danger), partially cutting & partially filling (balanced) – most economical.

❖Balancing depth

> Balancing depth = the depth of cutting at which the excavated earth just equals the earth required for forming the banks (embankment) – i.e. quantity of cutting = quantity of filling; gives minimum cost (no spoil bank or borrow pit).

- Depends on bed width, side slopes, bank top width, freeboard, FSL above ground. Found by trial/ using table or formula (e.g., for trapezoidal section: cut area = fill area).

❖Borrow pit

> Borrow pit = excavation (pit) made on the ground outside the canal to obtain earth for constructing the banks when excavation from the canal is insufficient (canal in filling). Should be dug on the canal-side (inside) of the land, not close to the bank toe (leave berm ≥ 3–5 m), kept shallow and drained.

❖Spoil bank

> Spoil bank = embankment formed from surplus excavated earth (spoil) when the canal is in deep cutting, deposited as mounds on one/ both sides near the canal (30 – 60 cm wide; berm ≥ 3 m). Use it as protective bund, plantation; placed only when excavation > filling. Must have gaps/ cross-sections for drainage & avoid blocking surface drainage.

❖Land width

> Land width = total width of land acquired for the canal (including bed, banks, berms, borrow pits, spoil banks, service roads, drains). Land width = bed width + 2 × (side slopes × depth) + 2 berms + 2 × bank top widths + extra for borrow pits/ spoil banks. Governed by the irrigation department's land-acquisition schedule.

5. Silt Theories – Introduction

- Canals carry silt (sediment) from the river; if velocity is too low → silting (deposition) reduces capacity; too high → scouring (erosion of bed & banks). Aim: design a non-silting, non-scouring (regime) channel.
- Regime channel: a channel that neither silts nor scours – stable cross-section & slope over years.
- Types of regime: initial regime (bed silt balanced – earth channels that attained stability), final (true) regime (all variables – discharge, silt charge, silt grade – constant; the channel is actually regime; practically only approached).
- Theories: Kennedy (1895, Upper Bari Doab canal); Lacey (1930s – regime theory); Gibbs, Blench, Garde etc.
- Silt supporting power – via eddies generated at the bed (vertical component of turbulence) keep silt in suspension.

6. Kennedy's Silt Theory

> Kennedy: "The silt in a channel is kept in suspension by the vertical components of the eddies generated by the friction of flowing water at the bed." The critical velocity V_0 is the mean velocity that makes the channel neither silt nor scour.

$V_0 = 0.55\ m\ D^{0.64}$ (V_0 in m/s; D = depth of water in channel, m)

- m = critical velocity ratio (C.V.R.) = V/V_0 – depends on silt grade: m = 1.0 for ordinary silt, 0.7–0.8 for very fine silt, 1.1–1.2 for coarse sandy silt. If m = 1 → no silting/ scour; m > 1 → scour; m < 1 → silt.
- Kutter's formula for velocity: $V = [(23 + 0.00155/S + 1/n) / (1 + (23 + 0.00155/S)\ n/\sqrt{R})]\sqrt{(RS)}$.
- Or Manning's: $V = (1/n)\ R^{2/3}\ S^{1/2}$ (modern).
- Also Kennedy's B/D ratio (bed width/ depth) taken from tables / by Garret's diagrams (graphical) to avoid trial (Garret).

7. Kennedy's Method of Channel Design

❖Given: discharge Q, silt factor m, rugosity n (0.0225), slope S (or B/D ratio).

❖Procedure – when S is known

- 1. Assume depth D (trial). 2. $V_0 = 0.55\ m\ D^{0.64}$. 3. Area $A = Q/V_0$. 4. Choose side slope (½:1 or 1:1 H:V); bed width B from $A = (B + D)\ D$. 5. Find $R = A/P$, wetted perimeter $P = B + 2.83\ D$ (side slope 1:1) or $B + 2.236\ D$ (½:1). 6. Compute $V = (1/n)\ R^{2/3}\ S^{1/2}$ (or Kutter). 7. If $V = V_0$ (within small difference) → design OK; else revise D and repeat.

❖Procedure – when B/D ratio is given

- 1. Assume D; $B = (B/D) \times D$, $A = (B + D)D$ (rect.) or $(B + zD)D$. 2. $V = Q/A$; 3. $V_0 = 0.55\ m\ D^{0.64}$; 4. If $V \neq V_0$, adjust D & repeat; 5. Then find slope S from Kutter/ Manning: $S = (V\ n / R^{2/3})^2$.

❖Example

> Design a channel for $Q = 10\ m^3/s$, m = 1, n = 0.0225, S = 1/5000, side slopes 1:1. Try D = 1.5 m: $V_0 = 0.55 \times 1.5^{0.64} = 0.55 \times 1.296 = 0.713$ m/s; $A = 10/0.713 = 14.03\ m^2$; $B = A/D - D = 9.35 - 1.5 = 7.85$ m; $P = B + 2.83\ D = 7.85 + 4.24 = 12.09$ m; $R = 14.03/12.09 = 1.16$ m; $V = (1/0.0225)\ 1.16^{2/3}\ (0.0002)^{1/2} = 44.4 \times 1.104 \times 0.01414 = 0.69\ m/s \approx V_0$ ✔ (accept; revise D to 1.47 for exact).

8. Drawbacks of Kennedy's Theory

- Silt grade treated only by empirical factor m; no definite relation with silt size (silt not defined); m is arbitrary.
- No equation for B/D or slope → trial & error (needs Garret's/ Lacey's supplement); B/D ratio arbitrary.
- Kutter's n chosen arbitrarily; no allowance for silt charge (quantity) or bed load.
- Based on only 22 channels of Upper Bari Doab canal (Punjab, a particular type of silt) – not general.
- Assumes silt is held by vertical eddies only – ignores concentration & water-silt interaction; considers bed width and depth independent; side slopes not studied.
- Ignores effect of sides (banks) on eddies – formula only for bed.
- Fails for very different silt, large canals.

9. Lacey's Regime Theory

Lacey: channel is in "regime" when it flows in incoherent alluvium of the same grade as that carried, with constant discharge & silt charge. Silt is held in suspension by the vertical components of eddies generated at ALL points of the wetted perimeter (bed & sides); there are three regimes – initial, final (true), and "in regime" conditions (final regime = cross-section & slope become constant, with the channel having a flow condition of semi-elliptical shape).

- Silt factor $f = 1.76\sqrt{d_{mm}}$ (d = mean particle size in mm; f = 1 for 0.55 mm; $f \approx 0.7–1.5$ for typical silt).

❖Lacey's regime equations (SI)

$$V = (Q f^2 / 140)^{1/6} \text{ (m/s)}$$

$$A = Q / V;\ P = 4.75\sqrt{Q} \text{ (wetted perimeter, m)}$$

$$R = 5 V^2 / (2 f) \text{ (hydraulic mean radius, m)}$$

$$S = f^{5/3} / (3340\, Q^{1/6}) \text{ (bed slope)}$$

- Other relations: $f = 1.76\sqrt{d}$, $V = 10.8\,R^{2/3}\,S^{1/3}$, $Q = A\,V$, $R = A/P$.
- Lacey's scour depth: $R_{scour} = 0.47\,(Q/f)^{1/3}$.

❖ **Lacey's design procedure (given Q and d (or f))**

- 1. Find silt factor $f = 1.76\sqrt{d_{mm}}$ (or given).
- 2. $V = (Q\,f^2/140)^{1/6}$.
- 3. $A = Q/V$.
- 4. $R = 5V^2/(2f)$.
- 5. $P = 4.75\sqrt{Q}$ (or $P = A/R$).
- 6. Using $P = B + 2.236\,D$ and $A = (B + 0.5\,D)\,D$ (side slope ½:1), solve for depth D and bed width B (quadratic).
- 7. $S = f^{5/3}/(3340\,Q^{1/6})$.

❖ **Example**

> Design by Lacey: $Q = 30\ m^3/s$, $d = 0.5$ mm. $f = 1.76\sqrt{0.5} = 1.245$. $V = (30\times1.245^2/140)^{1/6} = (0.3321)^{1/6} = 0.832$ m/s; $A = 30/0.832 = 36.06\ m^2$; $R = 5\times0.692/(2\times1.245) = 1.39$ m; $P = 4.75\sqrt{30} = 26.0$ m (also $P = A/R = 25.9$ ✔); $S = 1.245^{5/3}/(3340\times30^{1/6}) = 1.44/(3340\times1.763) = 1/4090 \approx 0.000245$.

10. Defects in Lacey's Theory

- Equations are empirical, derived from Indian & Egyptian canals (not universal); no rational base.
- Silt factor f is not a true property of silt; relation with d is empirical; concept of "regime" is ill-defined; true regime hardly ever reached.
- Silt charge (concentration) is ignored; only depends on Q & f; B/D is not independent (side slopes not accounted).
- Lacey's "eddy" concept (generated at all points) is unproven; hydraulic mean radius R relation doubtful; the sides of the channel treated as same as bed.
- $P = 4.75\sqrt{Q}$ relation shows P depends only on Q – regardless of bank soil & silt → inconsistent for stronger banks (rock/ cohesive).
- Not applicable to lined channels, rivers with bed load, very large/ small canals.

11. Comparison of Kennedy's and Lacey's Theories

Point	Kennedy	Lacey
Silt supported by	Vertical eddies from bed only	Eddies from bed and sides (whole perimeter)
Basic criterion	Critical velocity V_0	Regime velocity V
Silt grade	Critical velocity ratio m (arbitrary)	Silt factor $f = 1.76\sqrt{d}$ (from size)
Regime types	One (non-silting/ non-scouring)	Initial, final & true
Equations	One: $V_0 = 0.55\ m\ D^{0.64}$	Three (V, P, S) – no trial & error
Design	Trial & error (Garret's curves)	Direct with equations
Slope	Not given	Given
B/D ratio	Not given (assumed)	Obtained from regime equations
Based on data	Upper Bari Doab canals only	Many canals & rivers (more general)
Channel shape	Trapezoidal (any)	Semi-elliptical (final regime)
Silt charge	Not considered	Not considered

12. WATER LOGGING

> Water-logging = a condition in which the soil pores in the root zone become saturated with water due to rise of the water table (to within 1.5 m of ground surface in the root zone), so air circulation stops and plants suffer. (Capillary rise brings salts to surface.)

- Zones of water table: < 1.5 m – water-logged (danger); 1.5 – 3 m – critical (capillary-fringe effects); > 3 m – safe. Water-logged area in India ≈ 6 M ha (canal commands of Punjab, Haryana, Rajasthan, UP, A.P.).

❖Causes of water-logging

- Seepage from unlined canals & water-courses (main cause) and reservoirs/ tanks.
- Over-irrigation (excess application, flooding of fields), intensive irrigation without drainage.
- Inadequate natural drainage (flat topography, obstruction of natural drainage by canals, roads, railway embankments); blocking of drains.

- Impervious soil/ sub-soil layer (hard pan/ clay) near surface preventing percolation; high rainfall, flood; sea water intrusion/ high tide in coastal belts.
- Insufficient outlets/ lack of drainage design, un-planned canal alignment; poor irrigation practices (crop mixes needing little water); lack of pumping.

❖Effects of water-logging

- Depletion of oxygen in root zone → poor root growth, reduced yield, crop failure; anaerobic conditions → toxic substances; reduced soil temperature & bacterial activity (nitrification), delays tillage/ sowing.
- Salinity/ alkalinity (capillary rise deposits salts: "Reh/ Kallar/ Usar" lands); soil becomes sour (acidic/ alkaline).
- Breeding ground of mosquitoes & malaria, water-borne diseases; damage to buildings (dampness), roads, foundations; reduced stability of structures; loss of agricultural area & land value; ecological & economic loss; migration.
- Growth of weeds (e.g., Typha), waterlogged pasture.

❖Remedial (preventive & curative) measures

- Prevention: Reduce seepage – lining of canals & water courses; control of intensity of irrigation; efficient water management (proper depth & frequency, warabandi); proper crop rotation (deep-rooted & low-delta crops); avoid over-irrigation; optimum use of canal water with conjunctive use of ground water; adequate drainage design when planning projects; regulating supply to canals; maintain natural drainage (provide cross-drainage works for obstruction); limit reservoir seepage.
- Curative: Surface drainage – open drains; sub-surface (tile/ pipe) drainage; tube-well pumping (vertical drainage) to lower water table; improving natural drainage – deepening & straightening streams; tile drains/ mole drains; horizontal drains; re-sectioning/ pumping of ground water; biodrainage (Eucalyptus plantation); bunds & dikes.

❖Drainage systems (short)

- Surface drains (open ditches), sub-surface drains (perforated pipes in gravel envelope, spacing by Hooghoudt's equation), mole drains, vertical drainage via tube wells (skimming wells).
- Drain spacing (Hooghoudt): $L^2 = 8\,K\,d\,h\,/\,q + 4\,K\,h^2/q$.

13. Saline and Alkaline Soils & their Reclamation

❖Problem soils

- Saline soils (white alkali, solonchak): EC > 4 dS/m, ESP < 15%, pH < 8.5 – excess soluble salts (chlorides, sulphates of Na, Ca, Mg); white crust; reduce osmotic availability of water.
- Alkaline (sodic) soils (black alkali, solonetz): ESP (exchangeable sodium percentage) > 15%, pH > 8.5 (up to 10), EC < 4; dispersed clay, poor structure/ permeability, dark crust, sodium carbonate.
- Saline-alkaline soils: EC > 4 & ESP > 15.
- Causes: canal seepage & water-logging, high evaporation, brackish irrigation water, poor drainage, coastal sea-water, dry climate.

❖Reclamation

- Saline soils: leaching – apply excess good water (flooding) with drainage to wash salts below root zone; provide drainage (surface/ sub-surface); lower water table; salt-tolerant crops (barley, cotton, sugar beet, date palm); organic manure, mulching, green manure; scraping surface salts; deep ploughing; proper irrigation (frequent light irrigation).
- Alkali soils: chemical amendments to replace Na^+ with Ca^{2+}: gypsum ($CaSO_4 \cdot 2H_2O$) – most common; sulphur, sulphuric acid, pyrite, lime, calcium chloride, molasses, press mud; then leach with water; grow rice/ barley/ dhaincha (green manure), add organic matter.

$$2\text{Na-soil} + CaSO_4 \rightarrow \text{Ca-soil} + Na_2SO_4 \text{ (leached out)}$$

- Gypsum requirement (GR) = based on ESP reduction & CEC.
- Prevention: lined canals, proper drainage, conjunctive use, water-quality monitoring (SAR/ RSC), avoid over-irrigation.

14. Losses in Canals

> Canal water is lost between head and field by:

- Seepage (percolation) losses – through bed & sides; major (≈ 70–90% of total loss); depends on soil permeability, wetted perimeter, water depth, silt content, ground-water level, age of canal.

- Evaporation losses – from water surface (1–2%, ≈ 10–20 mm/day max in summer).
- Absorption (initial wetting) losses – dry canal bed soaks water at start of flow.
- Other: leakage, transpiration by weeds, operational wastage (spill, tail outfall), outlets leakage, gate & structure leakage.

❖Estimation of seepage

- Typical seepage rates (cumec per million m^2 of wetted area): sand 4–6; sandy loam 2.5–3.5; loam 1.5–2.5; clay 1–1.5; concrete-lined 0.3–0.6.
- Empirical formulas (Moritz, Punjab Irrigation Research Institute, Davis–Wilson) relate seepage to discharge, velocity, wetted perimeter and soil type; loss = rate × wetted area (P × L).
- Conveyance (transmission) loss ≈ 30–50% in unlined canals (A.P. average ≈ 30–40%).
- Control: lining, compaction, silt-laden water (natural clogging), reduce length, proper maintenance, bed-width/ depth ratio optimisation (narrow, deep sections lose less).

15. Lining of Irrigation Channels

> Canal lining = impervious (or semi-pervious) layer of concrete, brick, stone, asphalt, soil-cement, polythene/ geomembrane on bed & sides of canal to reduce seepage and erosion.

❖Necessity (why lining?)

- Reduce seepage loss (save water – 30–70%); control water-logging & salinity; prevent weeds; higher velocity allowed (non-scouring) → smaller cross-section & less land; lower maintenance (silting, weed control); protect banks (stability); safe against breaches; increase command area/ duty; more reliable supply, reduces cost of drainage; prevents rat holes.

❖Advantages

- Water saving (≈ 60–90% reduction in seepage); prevents water logging; smaller section → less land acquisition; less silting, no weed growth; permits high velocity (≈ 2 m/s) → smaller canal; lower maintenance cost; stable slopes (steeper side slopes); better discharge (smooth surface, n = 0.013–0.015 vs 0.025); less evaporation due to narrow deep section; more irrigation area per cumec; longer life & no breaches.

❖Disadvantages

- High initial cost (but pays back); repair difficult (cracks); problems of uplift pressure when canal is empty and ground water is high → failure (needs drainage/ pressure relief); not suitable for expansive soils/ frost without proper joints; hinders ground-water recharge; difficult to shift outlets; requires skilled labour & quality control; damage by animals/ people.

❖Types of lining

- (a) Hard-surface lining
  - Cement concrete lining – plain cement concrete (1:3:6 or 1:2:4), thickness 5 – 15 cm (bed & sides; slip-form paving) with expansion & contraction joints (every 3–6 m; filled with bitumen/ sealants) – most common. Also pre-cast concrete slabs/ tiles, shotcrete (gunite) with wire mesh for repairs, reinforced concrete for large canals.
  - Brick lining – single/ double layer of bricks laid in cement mortar over a bed of lime/ cement mortar; 7.5–15 cm; cheap in areas without stone; less durable.
  - Stone-masonry lining (dressed/ random rubble) – where stone is available; strong.
  - Asphaltic (bituminous) lining – asphalt concrete / bituminous membrane, flexible, tolerates settlement; for flexible soils.
- (b) Earth-type (soil) lining
  - Compacted earth lining (0.3–1.0 m of clayey soil), stabilised soil/ soil–cement (5–15% cement mixed with soil), bentonite/ clay blanket – cheap, but not durable against velocity.
- (c) Buried membrane lining – LDPE/ HDPE polythene film or geo-membrane (0.25 – 1 mm) covered by 15 cm soil/ concrete/ brick (protect from sunlight & puncture); cheap, flexible, quick – very popular for small canals.

- (d) Other: geosynthetic clay liner, ferro-cement, precast U-flumes, sprayed bitumen.

❖ **Choice of lining depends on**

- Cost, soil, canal size, velocity, seepage allowed, availability of material, ground-water level.

16. Design of a Lined Canal

Design is by Manning's formula, adopting hydraulically efficient (most economical) section (trapezoidal) with lined (smooth) surface and permissible velocity (non-silting & non-scouring, max 2.0–2.5 m/s for concrete lining).

❖ **Steps**

- 1. Given Q, bed slope S, Manning's n (0.015 for concrete lining; 0.016–0.02 brick), side slope z : 1 (1.5 : 1 for concrete; 1:1 stable soils), max velocity V_{max}.
- 2. Most economical trapezoidal section (best hydraulic section): half top width = side slope length, $R = D/2$; $B = 2D(\sqrt{(1+z^2)} - z)$.
- 3. Area $A = (B + zD)D$; wetted perimeter $P = B + 2D\sqrt{(1+z^2)}$; $R = A/P = D/2$.
- 4. $Q = AV = A(1/n)R^{2/3}S^{1/2}$ → solve D (trial or direct).
- 5. Check $V < V_{max}$; adopt freeboard (≈ 0.6 m for $Q \leq 10\ m^3/s$; 0.75 m up to 30 m^3/s), lining thickness, joints, berm width; design drainage/ pressure-relief valves under the lining.
- 6. Lining thickness t (cm) depends on size: 5 cm (small), 7.5 cm ($Q \leq 14$ cumec), 10–15 cm (> 28 cumec).

❖ **Example**

Design a most-economical concrete-lined trapezoidal canal: $Q = 15\ m^3/s$, $S = 1/5000$, $n = 0.015$, $z = 1.5$.

- Best section: $B = 2D(\sqrt{(1+2.25)} - 1.5) = 2D(1.803 - 1.5) = 0.606D$; $A = (0.606D + 1.5D)D = 2.106D^2$; $P = 0.606D + 2D \times 1.803 = 4.212D$; $R = D/2$.

- $Q = (1/0.015) \times 2.106D^2 \times (D/2)^{2/3} \times (0.0002)^{1/2} = 66.7 \times 2.106 \times 0.63 \times 0.01414 \times D^{8/3} = 1.251\, D^{8/3} = 15 \rightarrow D^{8/3} = 11.99 \rightarrow D = 2.53$ m; $B = 1.53$ m; $A = 13.5$ m^2; $V = 15/13.5 = 1.11$ m/s < 2.0 ✔.
- Provide freeboard 0.75 m; lining thickness 10 cm.

17. Solved Problems

❖Problem 1 – Kennedy's V_0

> $D = 2$ m, $m = 1.0$: $V_0 = 0.55 \times 2^{0.64} = 0.55 \times 1.558 = 0.857$ m/s.

❖Problem 2 – Lacey's

> $Q = 100$ m^3/s, $f = 1.0$: $V = (100/140)^{1/6} = 0.945$ m/s; $A = 105.8$ m^2; $P = 4.75 \times 10 = 47.5$ m; $R = 5 \times 0.893/2 = 2.23$ m; $S = 1/(3340 \times 100^{1/6}) = 1/(3340 \times 2.154) = 1/7195$.

❖Problem 3 – Seepage loss

> Wetted perimeter 20 m, length 5 km, loss 2.5 cumec/10^6 m^2: area = 10^5 m^2 → loss = 0.25 m^3/s.

❖Problem 4 – Saving by lining

> Unlined seepage = 3 cumec/million m^2; lined = 0.5; area = 2×10^6 m^2 → saving = $(3 - 0.5) \times 2 = 5$ m^3/s.

18. Important Questions – Unit IV

- Classify canals; canal alignment (ridge, contour, side-slope); inundation canals.
- Cross-section of an irrigation channel – balancing depth, borrow pit, spoil bank, land width.
- Kennedy's theory: assumptions, V_0 equation, design procedure & drawbacks.
- Lacey's regime theory: equations, design; defects; comparison with Kennedy.
- Water logging – causes, effects, remedies; saline and alkaline soils & reclamation.
- Canal losses; lining – necessity, advantages, disadvantages, types; design of lined canal.

Lacey: $V=(Qf^2/140)^{(1/6)}$; $P=4.75\sqrt{Q}$; $R=5V^2/2f$; $S=f^{(5/3)}/(3340 Q^{(1/6)})$; Kennedy: $V0=0.55$ m $D^{0.64}$